

(iii)

IN AL, 192

COH

Machine code is $\Rightarrow$

1110010
opcode

0
W

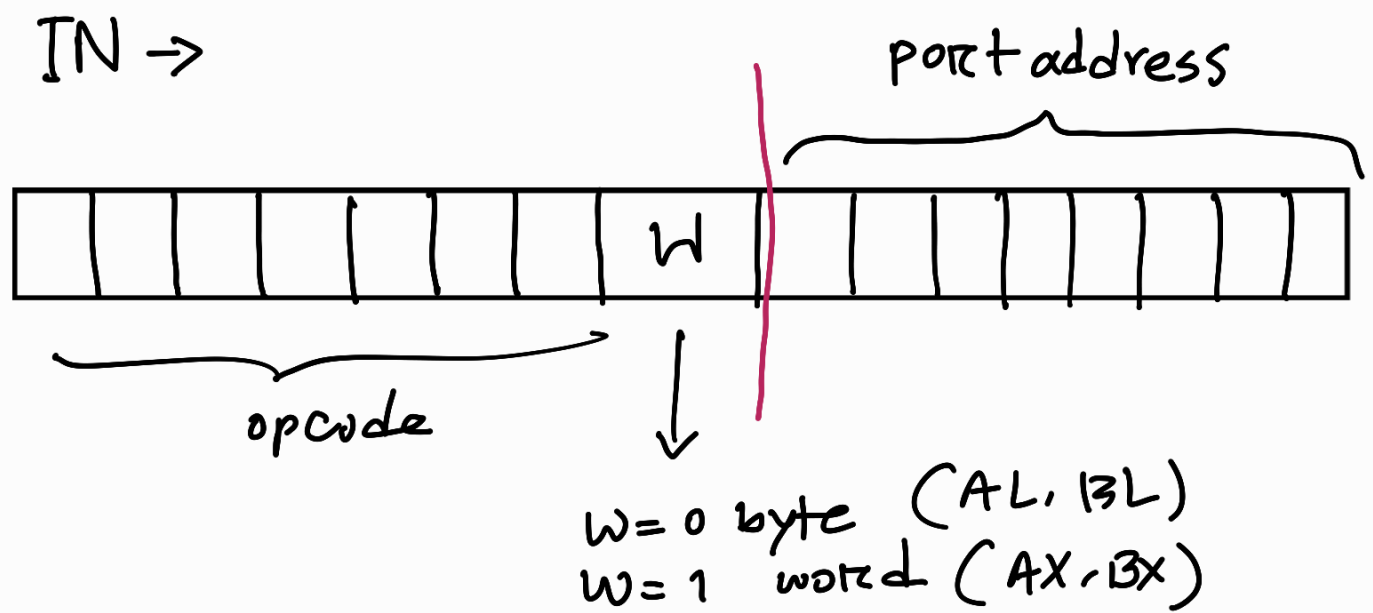
1100 0000
CO high byte 00
192 or COH

So, Machine code is $\Rightarrow$ E4 COH.

This is stored as:

ADDRESS	Content
00205	E4 H
00206	CO H

- Instructions have templates and coding formats
- Each have MOD and R/M bit patterns (specific)



IN AL, (8 bit port)

Opcode
E4 (1110 010 ^w0)

IN AL, 05H

Machine code → E4 05

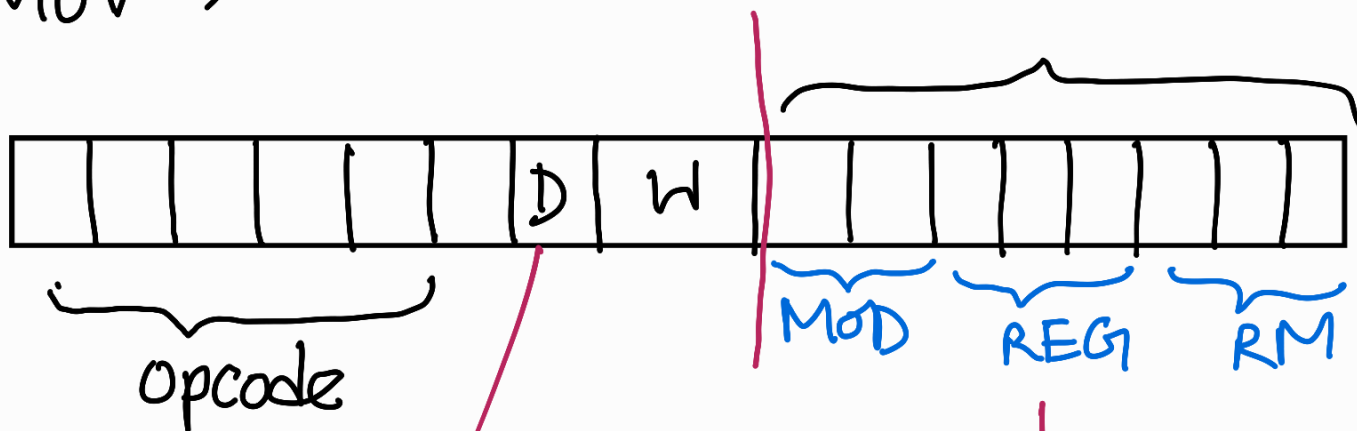
Address	Content
00205 H	E4 H
00206 H	05 H

IN AX, (8 bit port) → E5 (opcode) ^w
1110 0101

IN AL, DX → EC (1110 110⁰)

IN AX, DX → ED (1110 110¹)

MOV →



Direction
to / from register

Register
select

0 = from

1 = to

MOD & R/M

tell addressing mode

R/M → Register /
Memory

Opcode for MOV → 100010

* segment override prefix - 001XX110

ES → 00

CS → 01

SS → 10

DS → 11

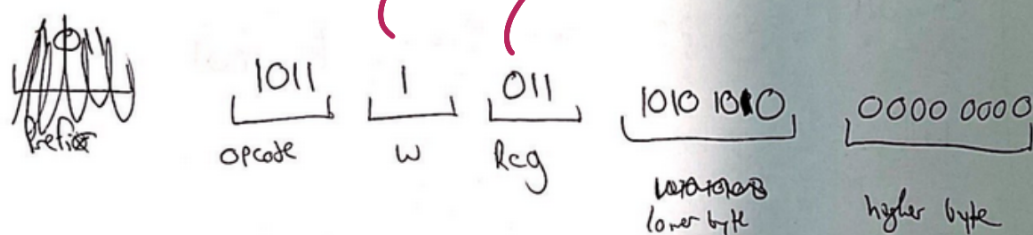
e.g.

MOV CS: [··], DX.

i

MOV BX, 1010 1010 B

So, machine code is,



So, the BBAA00H

full in HEX

lower
byte

higher
byte

In memory they may be stored as:

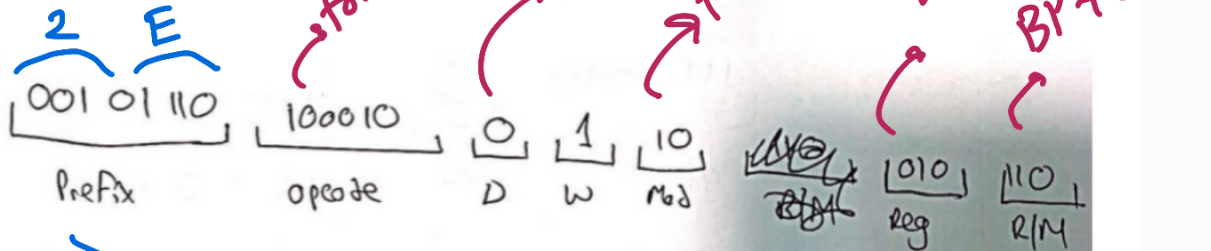
ADDRESS	CONTENT
00205	BB H
00206	AA H
00207	00 H

(ii)

code segment override
prefix: 2E

MOV CS: [BP + 1234H], DX.

Machine code:



in HEX →
2E8996

0011 0100 0001 0010
34H 12H

lower byte higher byte

So, this is 2E89963412H.

In memory it can be stored as:

Address	Contents
00205	2E H
00206	89 H
00207	96 H
00208	34 H
00209	12 H

- Q1. Derive the machine code for the instruction: **IN AL, F1 h**. Also, show how the derived contents can be stored in different memory locations. 06
- Q2. Your given Student Id format is 20 . . **XXXX**. Use last 4 (four) digits of your id as a hexa-decimal 09 value (i.e., **XXXXh**) and consider the following table.

RM	MOD	00		01		10		11	
								W = 0	W = 1
000		[BX] + [SI]		[BX] + [SI] + d8		[BX] + [SI] + d16		AL	AX
001		[BX] + [DI]		[BX] + [DI] + d8		[BX] + [DI] + d16		CL	CX
010		[BP] + [SI]		[BP] + [SI] + d8		[BP] + [SI] + d16		DL	DX
011		[BP] + [DI]		[BP] + [DI] + d8		[BP] + [DI] + d16		BL	BX
100		[SI]		[SI] + d8		[SI] + d16		AH	SP
101		[DI]		[DI] + d8		[DI] + d16		CH	BP
110		d16 (direct address)		[BP] + d8		[BP] + d16		DH	SI
111		[BX]		[BX] + d8		[BX] + d16		BH	DI

Now, using the table derive the machine code contents of the following MOV instructions and show how many memory locations are required to store the derived machine codes:

- MOV AL, [XXXXh]
- MOV SS:[BX+DI+ XXXXh], DX
- MOV BX, XXXXh

IN AL F1h

Machine code E4 F1

Address content

00205h E4

00206h F1

MOV AL, 1122h → AL is 8 bit, so invalid

MOV SS: [BX+DI+1122h], DX_{src}

stack segment
override - 36

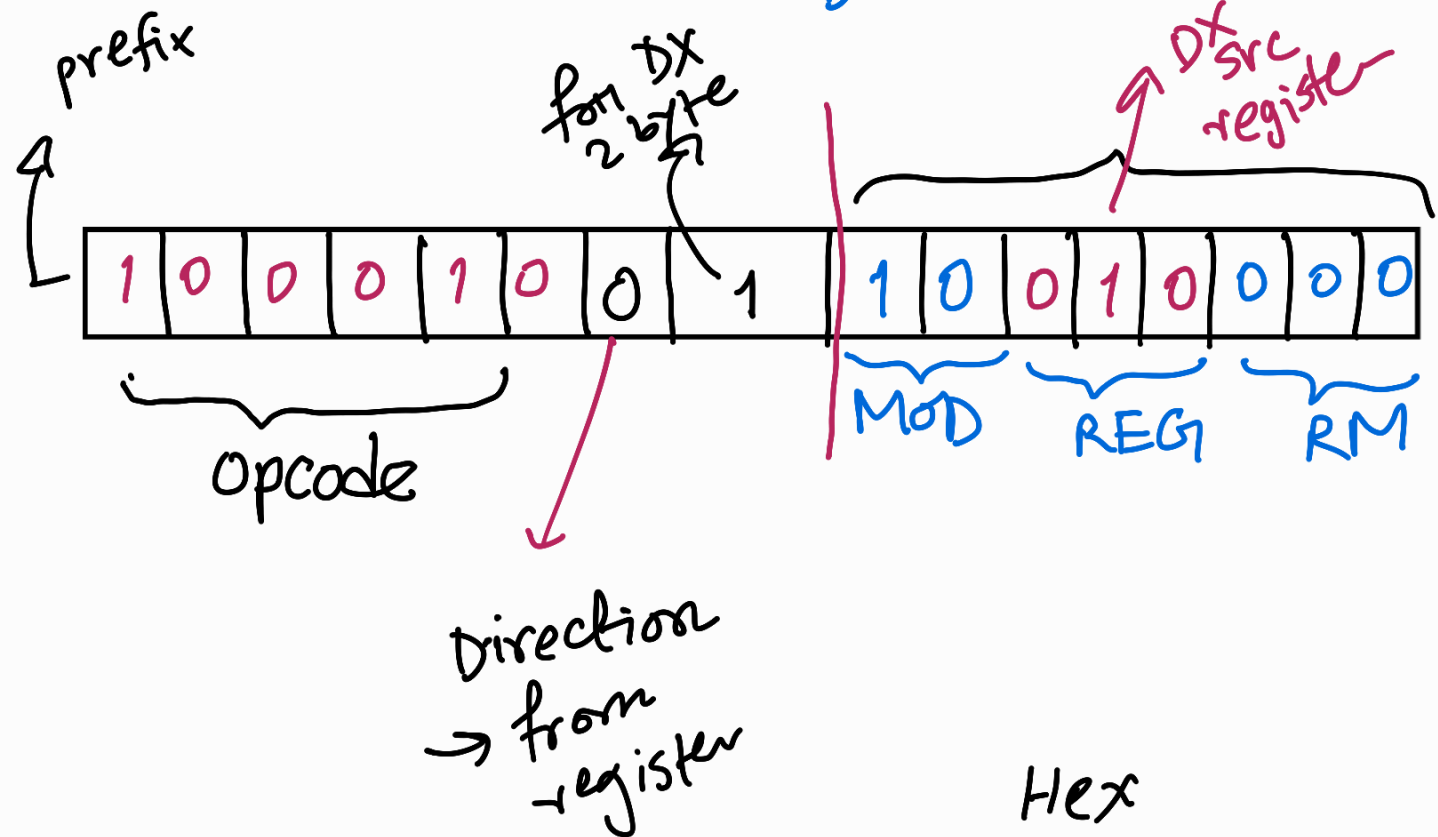
MOD → 10

REG → 010, W = 1

R/M → 000

0011 0110

D =



22 → low byte

11 → high byte

Hex

36h

89h

90h

22h

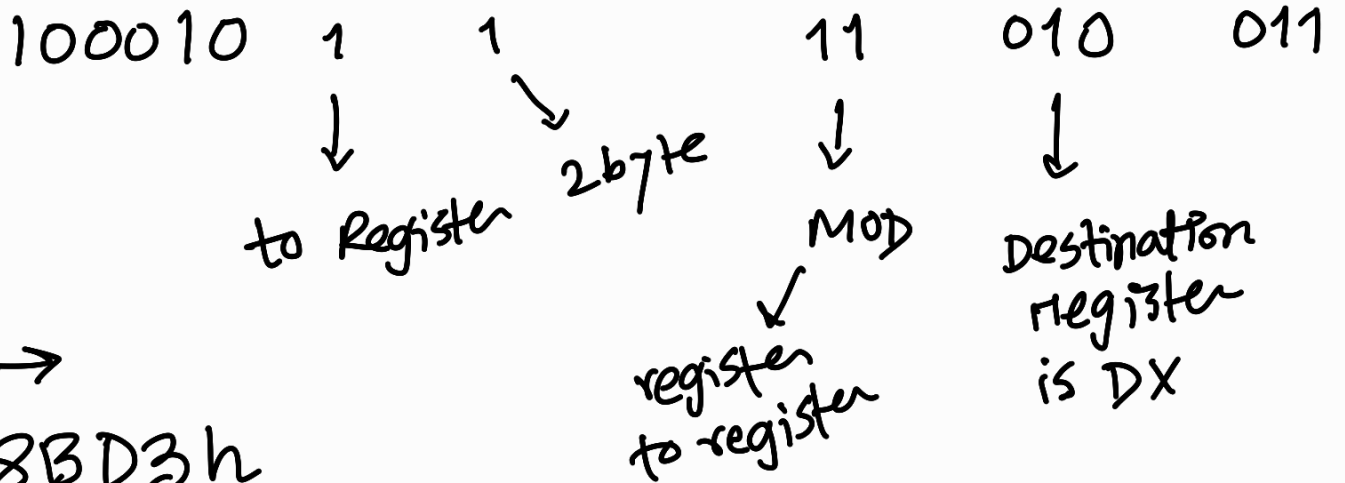
11h

5
memory
locations
needed

2. a) Derive the contents of the following MOV instructions using its coding template and also show how the contents of the instructions can be stored in memory: 12

- i. MOV DX, BX
- ii. MOV AAFh[DI], AH
- iii. MOV AX, [1234h]

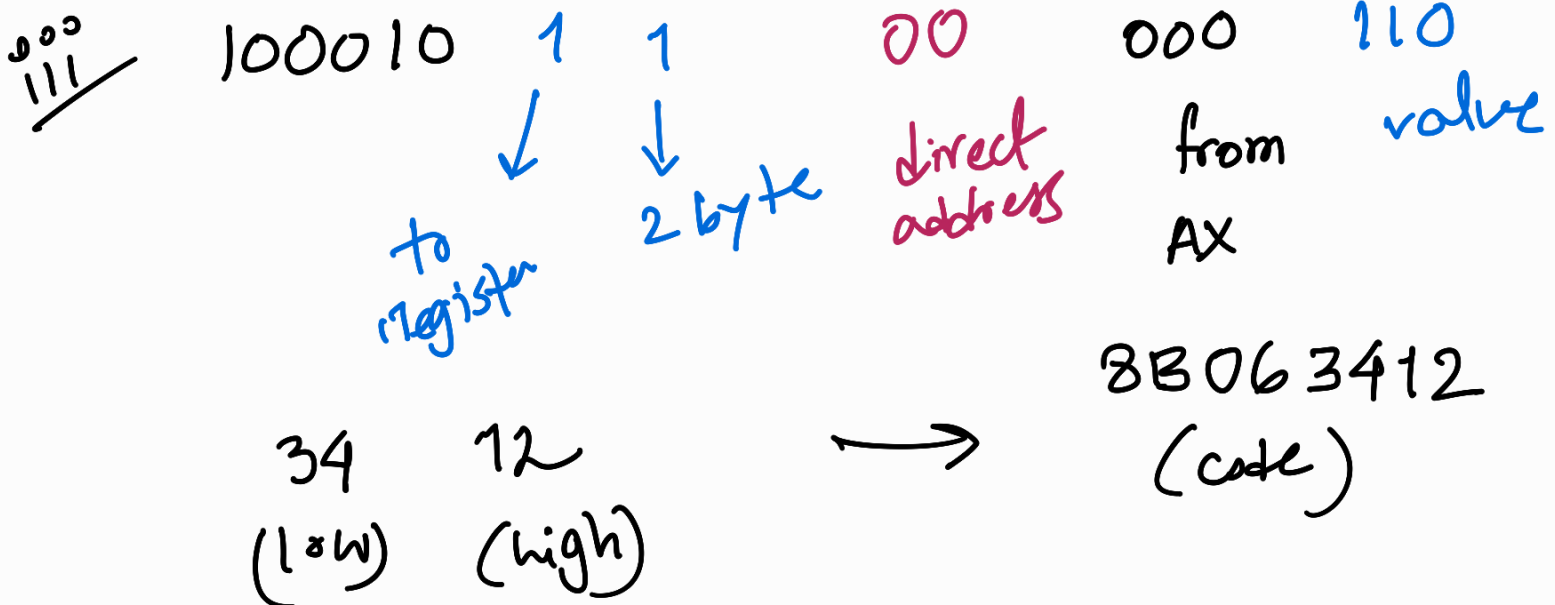
i / MOV DX, BX



Code →

8BD3h

Memory not used



ii MOV AAFh [DI], AH

16 bit displacement
MOD → 10

source register
AH REG = 100

RM → DI
W = 1
111

100010 0 0 10 100 111

from 1 byte

AAFFh

high low

Final code → 88 A7 FF AA

low high

b) Derive the machine codes of the following instructions using their instruction/coding template/format and also show how the machine codes of the instructions are to be stored in memory: (PO2)
3×4=12
(CO1)
(PO1)

- i. IN AL, 1234H
- ii. MOV SS:[BX+DI+A1B2H], DX
- iii. MOV BX, 01010101B

2. a) Derive the contents of the following MOV instructions using its coding template and also show how the contents of the instructions can be stored in memory: 12

- i. MOV AL, BL
- ii. MOV FFh[SI], BH
- iii. MOV DX, [ABCDh]

4. a) Derive the contents of the following MOV instructions using its coding template and also show how the contents of the instructions can be stored in memory. 10

- i. MOV DL, 00000001B
- ii. MOV SS:[SI], BP

c) Derive the contents of the IN AL, FFh using the instruction template and also show how the contents of this instruction can be stored in memory. 7

IN AL, FFh

E4 FF
mem1 mem2

3. a) Draw the coding template of IN instruction. Explain the significance of using 'MOD' and 'R/M' in MOV coding template. 9